

TRB-010 — Discovery Plan

Ticket: TRB-010

Request: Systematically identify invoices that have been submitted previously.

Type: Feature Request | **Priority:** 2 - Feature Request

1. What is Proposed

The arktaxrebate web application will be enhanced with a duplicate invoice detection system that activates during the invoice submission workflow. When a user submits one or more invoices, the system will compare each invoice against all previously submitted invoices on record for that taxpayer or submission context. The comparison will use a three-field composite key: Invoice Number, Invoice Date, and Invoice Amount. If all three fields match an existing submission record exactly, the invoice will be flagged as a probable duplicate before the submission is finalized.

The implementation consists of the following components:

- **Backend duplicate-check service:** A new server-side function or API endpoint that accepts the incoming invoice payload and queries the invoice records store for any existing entry where Invoice Number, Invoice Date, and Invoice Amount all match. The check runs before the submission is committed.
 - **Composite key matching logic:** Matching will be case-insensitive and whitespace-normalized for Invoice Number, exact-date matching (normalized to ISO 8601) for Invoice Date, and exact numeric equality (rounded to two decimal places) for Invoice Amount.
 - **Duplicate flag on the data model:** The invoice data model will gain a `duplicateOf` reference field (nullable) that, when populated, stores the ID of the previously submitted invoice it matches. Flagged invoices will not be auto-rejected; they will surface to the user for review.
 - **Frontend warning UI:** During the submission review step, any flagged invoice will render an inline alert stating the invoice appears to match a previously submitted invoice, including the prior submission date and submission ID. The user is presented with two explicit actions: **Remove this invoice** or **Keep and submit anyway** (requires acknowledgment click).
 - **Audit logging:** When a user overrides a duplicate flag, the override decision, timestamp, and user identity are written to the audit log.
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2. Desired Outcome

When this work is complete, no duplicate invoice will pass through the submission workflow without the submitting user being explicitly informed and required to make a deliberate choice. Users who accidentally include a re-submitted invoice will see a clear, actionable warning at the point of submission rather than discovering the problem after the fact. The system will reduce the volume of duplicate invoices reaching

the review queue, decrease the manual workload on portal administrators, and provide an auditable record of every case where a user knowingly proceeded with a potential duplicate.

3. Success Criteria

- When an invoice is submitted whose Invoice Number, Invoice Date, and Invoice Amount all exactly match a previously submitted invoice, the submission workflow halts at the review step and displays a duplicate warning identifying the prior submission.
- When any one of the three fields differs from all existing records, no duplicate warning is shown and the submission proceeds normally.
- The duplicate warning UI displays the prior submission date and reference ID so the user has enough context to make an informed decision.
- Clicking "Remove this invoice" removes the flagged invoice from the current batch; remaining invoices can still be submitted without interruption.
- Clicking "Keep and submit anyway" requires an explicit confirmation action before the submission proceeds.
- When a user overrides a duplicate flag, the audit log records the user identity, invoice identifier, matching prior submission ID, override decision, and timestamp.
- Invoice Number matching is case-insensitive and trims leading/trailing whitespace.
- Invoice Date matching normalizes both dates to ISO 8601 before comparison.
- Invoice Amount matching compares values rounded to two decimal places.
- If a batch contains intra-batch duplicates, each is individually flagged.
- The duplicate check completes in under 3 seconds for a batch of up to 500 invoices.
- Existing prior submissions are included in the historical comparison set so the feature is retroactively effective.